

S21 Full Hyperparameter Search Spaces and Selection Logs

Purpose

This supplementary section documents all explored hyperparameter ranges and records the final selected configurations for each evaluated model to ensure full experimental reproducibility and methodological transparency.

1. HIMS-CDI (Multinomial Logistic Regression)

Search Space

- Regularisation strength: $C \in \{0.01, 0.1, 1, 10, 100\}$
- Solver: lbfgs, saga
- Maximum iterations: 100, 500, 1000

Selection Procedure

Five-fold cross-validation was performed on the training partition (60% of the 240 synthetic environments) using macro-averaged F1-score as the optimisation criterion.

Selected Configuration

$C = 1.0$
solver = lbfgs
max_iter = 1000

Justification

The selected configuration achieved the highest macro-F1 score with minimal calibration error.

2. Graph Neural Network (GNN)

Search Space

- Hidden dimensions: {32, 64, 128}
- Learning rate: $\{1 \times 10^{-4}, 1 \times 10^{-3}, 1 \times 10^{-2}\}$
- Dropout rate: {0.2, 0.5}
- Number of layers: 2, 3
- Optimiser: Adam, AdamW

Selection Procedure

Random search across 20 parameter combinations with early stopping (patience = 10 epochs).

Selected Configuration

Hidden dimension = 64
Learning rate = 1×10^{-3}
Dropout = 0.5
Layers = 2

Optimiser = AdamW
Weight decay = 1×10^{-4}

Validation Log

Best validation loss: 0.297 at epoch 39 (seed 13).

3. Transformer

Search Space

- Hidden size: {256, 512}
- Attention heads: {4, 8}
- Layers: {2, 3, 4}
- Learning rate: $\{1 \times 10^{-5}, 3 \times 10^{-5}, 1 \times 10^{-4}\}$
- Dropout: {0.1, 0.2}
- Optimiser: AdamW, RAdam

Selected Configuration

Hidden size = 512
Heads = 8
Layers = 3
Learning rate = 3×10^{-5}
Dropout = 0.1
Optimiser = RAdam

Validation Log

Best validation loss: 0.261 (seed 73).

4. Edge-Aware Fusion

Search Space

- Compression ratio: {0.3, 0.5, 0.7}
- Learning rate: $\{1 \times 10^{-4}, 1 \times 10^{-3}\}$
- Batch size: {16, 32, 64}
- Optimiser: SGD, Adam

Selected Configuration

Compression ratio = 0.5
Learning rate = 1×10^{-3}
Batch size = 32
Optimiser = Adam

Validation Log

Best validation loss: 0.324 (seed 13).

5. Rule-Based and Checklist Models

No hyperparameter optimisation was required.

6. Equal Weights Composite

No parameter tuning was required because all dimensions were weighted equally.

Reproducibility

All hyperparameter searches were conducted using fixed random seeds. Configuration files are archived within:

`/configs/hyperparameter_search/`

The selected parameters are additionally embedded within:

`reproduce_all.py`

thereby enabling full experimental replication.